SE Lab - 7 Fuzz Testing

Name - Tanvi. B. Shetty
SRN - PES1UG24AM302
Section - F

1.

```
C:\Users\Tanvi\Desktop\SEM - 5\SE\Lab 7\L6_L7_02_car_rental_rate\Lab 7>python processor.py
==== Running sanitize_license_plate ====
Enter license plate with delimiters (-, _, #): KA-09_AB#1234
Sanitized Plate: KA09AB1234

==== Running parse_trip_mileage ====
Enter CSV of mileages (e.g. 100.5, 45.2, 80.0): 98.2
Parsed Mileage List: [98.2]

==== Running format_rental_invoice ====
Enter rental record (Driver:Days:Category): Manish:10:SUV
Rental Invoice: Driver Manish booked SUV for 10 days: Total $500.00
```

2.

```
C:\Users\Tanvi\Desktop\SEM - 5\SE\Lab 7\L6_L7_02_car_rental_rate\Lab 7>pytest test_processor.py
============================= test session starts ==============================
platform win32 -- Python 3.13.5, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Tanvi\Desktop\SEM - 5\SE\Lab 7\L6_L7_02_car_rental_rate\Lab 7
plugins: anyio-4.11.0, hypothesis-6.170.15, langsmith-0.4.37
collected 3 items

test_processor.py FFF                                                    [100%]

================================== FAILURES ===================================
_____________________ test_sanitize_license_plate_no_crash _____________________

s = None

    @given(st.text() | st.none())
    def test_sanitize_license_plate_no_crash(s):
        try:
>           sanitize_license_plate(s)

test_processor.py:7:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

plate = None

    def sanitize_license_plate(plate):
>       plate = plate.strip()
                ^^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

processor.py:2: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_sanitize_license_plate_no_crash(s):
                ^^^

test_processor.py:5:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

s = None

    @given(st.text() | st.none())
    def test_sanitize_license_plate_no_crash(s):
        try:
            sanitize_license_plate(s)
        except Exception as e:
>           assert False, f"sanitize_license_plate crashed with: {e!r}"
E           AssertionError: sanitize_license_plate crashed with: AttributeError("'NoneType' object has no attribute 'strip'")
E           assert False
```

```
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Tanvi\Desktop\SEM - 5\SE\Lab 7\L6_L7_02_car_rental_rate\Lab 7\test_processor.py", line 23, in test_format_rental_invoice_safe
    |     assert False, f"format_rental_invoice crashed with: {e!r}"
    | AssertionError: format_rental_invoice crashed with: IndexError('list index out of range')
    | assert False
    | Failing test case: test_format_rental_invoice_safe(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Tanvi\Desktop\SEM - 5\SE\Lab 7\L6_L7_02_car_rental_rate\Lab 7\test_processor.py", line 21, in test_format_rental_invoice_safe
    |     format_rental_invoice(s)
    |     ^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Tanvi\Desktop\SEM - 5\SE\Lab 7\L6_L7_02_car_rental_rate\Lab 7\processor.py", line 12, in format_rental_invoice
    |     parts = record.split(':')
    |             ^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Tanvi\Desktop\SEM - 5\SE\Lab 7\L6_L7_02_car_rental_rate\Lab 7\test_processor.py", line 23, in test_format_rental_invoice_safe
    |     assert False, f"format_rental_invoice crashed with: {e!r}"
    | AssertionError: format_rental_invoice crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_format_rental_invoice_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=========================== short test summary info ===========================
FAILED test_processor.py::test_sanitize_license_plate_no_crash - AssertionError: sanitize_license_plate crashed with: AttributeError("'NoneType' object has no attribute 'strip'")
FAILED test_processor.py::test_parse_trip_mileage_safe - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_processor.py::test_format_rental_invoice_safe - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 2.60s ==============================
```

3.

```
C:\Users\Tanvi\Desktop\SEM - 5\SE\Lab 7\L6_L7_02_car_rental_rate\Lab 7>python processor.py
==== Running sanitize_license_plate ====
Enter license plate with delimiters (-, _, #): ka-09_ab#1234
Sanitized Plate: KA09AB1234

==== Running parse_trip_mileage ====
Enter CSV of mileages (e.g. 100.5, 45.2, 80.0): 100.5,,abc,45.2
Parsed Mileage List: [100.5, 45.2]

==== Running format_rental_invoice ====
Enter rental record (Driver:Days:Category): Manish:abc:SUV
Rental Invoice: Invalid record
```

4.

```
C:\Users\Tanvi\Desktop\SEM - 5\SE\Lab 7\L6_L7_02_car_rental_rate\Lab 7>pytest test_processor.py
============================= test session starts =============================
platform win32 -- Python 3.13.5, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Tanvi\Desktop\SEM - 5\SE\Lab 7\L6_L7_02_car_rental_rate\Lab 7
plugins: anyio-4.11.0, hypothesis-6.170.15, langsmith-0.4.37
collected 3 items

test_processor.py ...                                                    [100%]

============================== 3 passed in 0.58s ==============================
```